Assignment #1: Fingerprint Recognition

Aleksandar Mijatović
IBB 2026/27 , FRI, UL
am93067@student.uni-lj.si

I. Introduction

Our goal is to perform and evaluate fingerprint recognition with the NIST Biometric Image Software (NBIS): to select the *bozorth3* score threshold that best separates genuine (same finger) from impostor (different finger) comparisons, to classify pairs of fingerprints with it, and to test how restricting the matching to the pattern type predicted by *pcasys* changes the result.

II. Methodology

Matching. The images were converted from TIFF to PNG, *mindtct* extracted the minutiae and *bozorth3* compared all images with each other. Its score is not exactly symmetric, so both A vs. B and B vs. A were computed, only self-comparisons were skipped. A comparison is genuine if both images show the same finger (from the file name) and impostor otherwise. Image quality was rated with *nfiq* (1 = best, 5 = worst).

Threshold. A pair is accepted as the same finger if its score is at least t. We compared (i) Otsu's method on all scores (own implementation and scikit-image), (ii) the equal error rate (EER), where the false accept rate (FAR) equals the false reject rate (FRR), and (iii) an accuracy sweep over all integer thresholds. We use the sweep (the lowest threshold if several tie), since accuracy is what we maximize. The whole dataset is used for selection and evaluation.

Classification. Every pair is compared only once (no A vs. A, no repeated B vs. A) and the prediction (score $\geq t$) is checked against the file-name labels. In the second experiment *pcasys* first assigns each image (converted to WSQ) a pattern type (A, L, R, S, T, W), and *bozorth3* is used only for pairs of the same type, all other pairs are labelled impostors.

III. Experiments

We used FVC2002 DB1_B: 10 fingers with 8 impressions each (80 images), and NBIS 5.0.0. All-vs-all matching gives 6320 comparisons (560 genuine, 5760 impostor) for the plots and the threshold, the classification uses 3160 unique pairs (280 genuine, 2880 impostor). Otsu's method used 256 histogram bins.

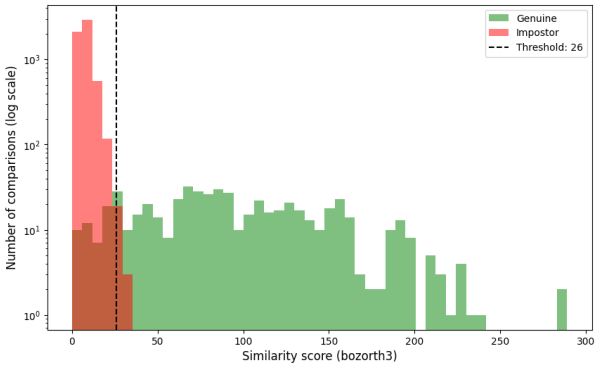


Fig. 1. *bozorth3* scores of genuine (green) and impostor (red) comparisons, log scale. Dashed line: selected threshold $t = 26$.

TABLE I
Threshold selection methods on all 6320 comparisons.

Method	t	Acc. [%]	FAR [%]	FRR [%]
Otsu (own)	65.5	97.25	0.00	31.07
Otsu (skimage)	64.9	97.37	0.00	29.64
EER	16	95.92	4.06	4.29
Accuracy sweep	26	98.92	0.21	10.00

IV. Results and Discussion

We show the score distributions, the threshold selection and the two classification experiments.

A. Results

Impostor scores are low (median 7, maximum 33) while genuine scores spread widely (median 88, maximum 289), see Fig. 1. The similarity matrix (Fig. 2) has ten bright diagonal blocks, the genuine comparisons of each finger, and is not symmetric. NFIQ (Fig. 3) rates all images between 1 (best) and 3, none is worse.

The accuracy sweep reaches its maximum (98.92%) for every t from 26 to 28, we take $t = 26$ (Table I). Otsu's method ($t \approx 65$) rejects about 30% of the genuine comparisons and the EER ($t = 16$) gives a lower accuracy.

Table II gives the classification at $t = 26$: the accuracy is **98.89%** (3125 of 3160 pairs) with *bozorth3* alone and **97.94%** (3095 of 3160 pairs) with *pcasys* first, which needs 1179 instead of 3160 comparisons (62.7% fewer).

TABLE II
CLASSIFICATION OF THE 3160 UNIQUE PAIRS AT $t = 26$. CALLS: NUMBER OF *BOZORTH3* COMPARISONS.

Experiment	Calls	Acc. [%]	FAR [%]	FRR [%]
bozorth3	3160	98.89	0.24	10.00
pcasys + *bozorth3*	1179	97.94	0.14	21.79

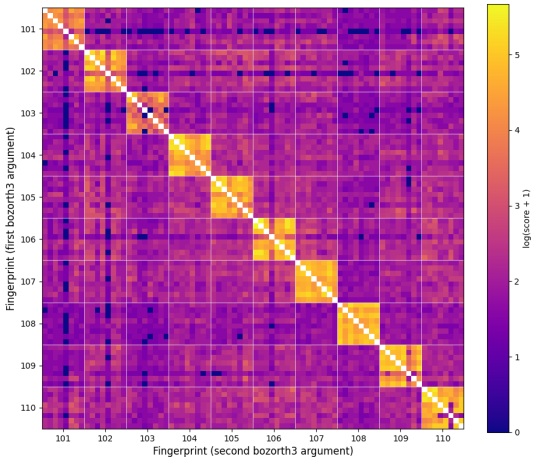


Fig. 2. Similarity matrix of all comparisons, log(score + 1). Rows: first, columns: second *bozorth3* argument. White diagonal: skipped self-matches.

V. CONCLUSION

The threshold $t = 26$ from the accuracy sweep classifies 98.89% of the 3160 pairs correctly, better than the Otsu and EER thresholds. Matching only within the *pcasys* type saves 62.7% of the comparisons but lowers the accuracy to 97.94%. Threshold selection and evaluation share one small dataset, so the accuracies are optimistic.

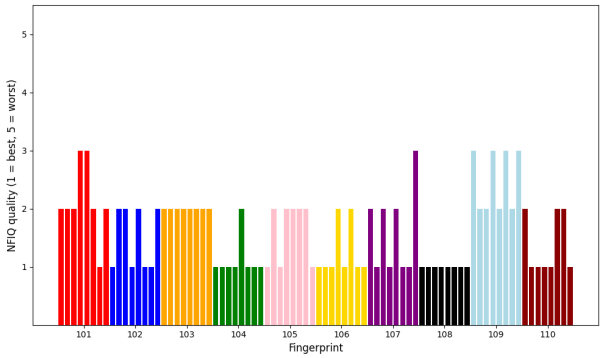


Fig. 3. NFIQ quality of the 80 images (1 = best, 5 = worst), one colour per finger.